Sara Mohamed Mouhanad Al-Sati

AI Engineer | LLM Agents & RAG | Machine Learning | Backend Development

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PROFESSIONAL SUMMARY

Artificial Intelligence and Computer Science graduate with hands-on experience in Machine Learning, Deep Learning, Generative AI, RAG systems, tool-calling LLM agents, backend development, and data analytics. Experienced with Python, FastAPI, LangChain, vector search, prompt engineering, and machine learning algorithms, plus backend development with PHP, Laravel, REST APIs, MySQL, authentication, and authorization. Built end-to-end AI applications integrating models, APIs, databases, retrieval systems, and web interfaces.

EDUCATION

B.Sc. Computer Science & Artificial Intelligence

Arab Open University – Egypt (2022–2026) | Major: Artificial Intelligence

PROJECTS

Enterprise AI Knowledge Assistant – RAG System

Python | FastAPI | RAG | LangChain | LLM | Hybrid Search | React

- Developed an enterprise RAG application for answering questions from internal company documents.
- Built FastAPI services for authentication, document ingestion, retrieval, and LLM-powered question answering.
- Implemented document parsing, cleaning, chunking, metadata extraction, indexing, and source attribution for PDF, DOCX, TXT, and Markdown.
- Implemented hybrid retrieval combining semantic/vector search with BM25 and metadata-based department access control.
- Implemented JWT authentication and role-based authorization for employees and administrators.
- Built a React frontend and applied production-oriented security, retrieval quality, prompt-injection awareness, and evaluation concepts.

Customer Support AI Agent

Python | LLM Tool Calling | Gemini / Groq (OpenAI-compatible API) | Streamlit

- Built an autonomous LLM agent that reads customer issues and decides which tools to call and in what order (order lookup, shipping check, cancel, refund, ticket creation), with no hard-coded routing logic.
- Implemented a custom agent loop (LLM → tool → result → LLM, max 8 steps) with conversation memory, multi-step tool chaining, and human handoff via support tickets when business rules are triggered (e.g., delivery delays, high-value refunds).
- Designed reliability features: tool errors returned to the model, automatic retries for transient tool failures and LLM rate limits/5xx errors, and a guardrail that blocks replies claiming an action (e.g., ticket created) that was never executed.
- Built a Streamlit chat UI with a tool-call inspector for traceability; validated behavior across 5 end-to-end scenarios (handoff, cancellation, refund, memory, retry-on-failure).

Diabetes Risk Prediction and Tracking System (DRPTS)

Python | XGBoost | Flask | Laravel | MySQL | SHAP | Machine Learning

- Developed an end-to-end AI web application for non-invasive diabetes and prediabetes risk assessment.
- Built preprocessing, feature engineering, class balancing, model training, and evaluation pipelines; trained an XGBoost classifier and applied SMOTE to address class imbalance.
- Integrated a Flask prediction API with a Laravel web application and used SHAP for model explainability.
- Developed risk classification, probability visualization, prediction history, dashboards, personalized recommendations, and an LLM-powered assistant.

NovaCart – E-Commerce Platform

Laravel | PHP | MySQL | REST API | Filament

- Developed a Laravel e-commerce backend with product, category, order, user, and supplier management.
- Implemented CRUD, Eloquent relationships, validation, pagination, search, filtering, authentication, and authorization policies.
- Built REST APIs for products, categories, and guest checkout; implemented image uploads with automatic file deletion.

Machine Learning, Deep Learning & BI Projects

Python | SQL | Power BI | Tableau | Excel

- Developed classification, regression, clustering, and model evaluation solutions with preprocessing and feature engineering pipelines; worked with neural networks, CNNs for computer vision, and NLP approaches, applying hyperparameter optimization.
- Developed interactive dashboards and analytical reports; used SQL for data extraction and analysis; performed data cleaning, modeling, exploratory analysis, and KPI development on real-world datasets.

TECHNICAL SKILLS

Programming: Python, PHP, SQL, C++

AI/ML: Machine Learning, Deep Learning, Feature Engineering, Data Preprocessing, Model Evaluation, XGBoost, Neural Networks, CNN, RNN, NLP

Generative AI: LLMs, Prompt Engineering, Context Engineering, RAG, LangChain, AI Agents, Tool Calling, Agentic Workflows, MCP, Multi-Agent Systems, LLM Evaluation, Guardrails, Hugging Face

RAG/IR: Document Processing, Chunking, Metadata Filtering, Vector Search, BM25, Hybrid Search, Reranking, Semantic Search, Multi-Tenant Retrieval

Backend: FastAPI, Flask, Laravel, REST APIs, Authentication, Authorization, File Uploads, Validation

Databases: MySQL, SQL Server, Eloquent ORM, Database Design

Data/BI: EDA, Data Cleaning, Data Visualization, Power BI, Tableau, Excel

Tools: React, JavaScript, Streamlit, Git, GitHub, Docker, Jupyter, VS Code

PROFESSIONAL TRAINING

Phronesis Engineering Cohort – Generative AI Track: LLM Applications, Context Engineering, RAG, AI Agents, MCP, Multi-Agent Systems, Production AI, Guardrails, Prompt Injection, LLM Evaluation, Deployment

Business Intelligence Track – Information Technology Institute (ITI), 2025: Data Warehousing, SQL Server, Power BI, Tableau, Excel for Data Analysis, Statistics Fundamentals

Front-End Web Developer Diploma – Training Support Unit, Arab Open University

CERTIFICATIONS

- Machine Learning Specialization – DeepLearning.AI | Andrew Ng | Coursera
- Deep Learning Specialization – DeepLearning.AI | Andrew Ng | Coursera
- CS50 – Harvard / edX
- Business Intelligence Track – Information Technology Institute (ITI)

LEADERSHIP & VOLUNTEERING

Head of HR – Google Developer Student Clubs (GDSC), Arab Open University

- Managed recruitment and onboarding; coordinated internal team operations and member activities.
- Organized technical workshops and events and collaborated across teams to support student development.

ADDITIONAL SKILLS

Analytical Thinking | Problem Solving | Communication | Teamwork | Time Management | Fast Learning | Technical Research | Continuous Learning